Indian Institute of Information Technology Bhopal

Computer Science & Engineering (Data Science)

Name of Program	B. Tech. CSE (DS)	**Semester-** Fifth	**Year-**Third
Course Name	Optimization Techniques		
Course Code	CSE(DS)-5003		
Compulsory/Elective/Open Elective	Elective		

Prerequisites

Data Mining and Warehousing (CSE(AI)-4004)

Course Learning Objectives

1. To gain knowledge on theory of optimization and conditions for optimality for unconstraint and constraint optimization problems.
2. To gain knowledge on theory of optimization and conditions for optimality for unconstraint and constraint optimization problems.
3. To understand programming techniques and implement different optimization techniques to solve various models arising from engineering areas.

Course Content

Module 1: Linear Programming (L.P): Revised Simplex Method, Dual simplex Method, Sensitivity Analysis Dynamic Programming (D.P): Multistage decision processes. Concepts of sub optimization, Recursive Relation-calculus method, tabular method, LP as a case of D.P.

Module 2: Classical Optimization Techniques: Single variable optimization without constraints, Multi variable optimization without constraints, multivariable optimization with constraints – method of Lagrange multipliers, Kuhn-Tucker conditions. Numerical Methods For Optimization: Nelder Mead's Simplex search method, Gradient of a function, Steepest descent method, Newton's method.

Module 3: Modern Methods of Optimization: Genetic Algorithm (GA): Differences and similarities between conventional and evolutionary algorithms, working principle, Genetic Operators- reproduction, crossover, mutation Genetic Programming (GP): Principles of genetic programming, terminal sets, functional sets, differences between GA & GP, Random population generation. Fuzzy Systems: Fuzzy set Theory, Optimization of Fuzzy systems.

Module 4: Integer Programming: Graphical Representation, Gomory's Cutting Plane Method, Balas' Algorithm for Zero–One Programming, Branch-and-Bound Method. Applications Of Optimization in Design and Manufacturing Systems: Formulation of model- optimization of path synthesis of a four-bar mechanism, minimization of weight of a cantilever beam, general optimization model of a machining process, optimization of arc welding parameters, and general procedure in optimizing machining operations sequence.

Course Outcomes
At the end of the course students will be able to: CO 1: Formulate the Engineering Problems as an Optimization Problem. CO 2: Use classical optimization techniques and numerical methods of optimization. CO 3: Justify and apply the use of modern heuristic algorithms for solving optimization problems. CO 4: Enumerate fundamentals of Integer programming technique and apply different techniques to solve various optimization problems arising from engineering areas.
List of Text Books
1. Optimization for Engineering Design by Kalyanmoy Deb, PHI Publishers, 2012. 2. Genetic algorithms in Search, Optimization, and Machine learning – D.E.Goldberg, Addison-Wesley Publishers.
List of Reference Books
1. Operations Research by Hillar and Liberman, TMH Publishers. 2. Optimal design – Jasbir Arora, Mc Graw Hill (International) Publishers.